

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Vectors

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

VECTORS · CH5 · L1

ENG. ESLAM AHMED



The Map

This lesson collapses Vectors into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Follow the route

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use equal parallel vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use midpoint or ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Prove collinearity

TRIGGER *"show that", "prove", "give a reason"*

05 Use vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 41 website questions, 41 checked solutions, 4 local PDFs read.

01 Follow the route

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\overrightarrow{OA} = \mathbf{a}, \quad \overrightarrow{OB} = \mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q14

Use vectors

$$\overrightarrow{OA} = \mathbf{a}, \quad \overrightarrow{OB} = \mathbf{b}$$

(a)(i)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}$$

Finish: (i) $\mathbf{b} - \mathbf{a}$, (ii) $4\mathbf{b} - 2\mathbf{a}$, (iii) $2\mathbf{b} - 3\mathbf{a}$, (b) 90 cm^2 .

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

14 ABCDEF and GHIJKL are regular hexagons each with centre O. Diagram NOT accurately drawn G L I H K J E D A B F C O a b GHIJKL is an enlargement of ABCDEF, with centre O and scale factor 2 OA to = a OB to = b (a) Write the following vectors, in terms of a and b. Simplify your answers. (i) AB to (1) (ii) KI to (2) (iii) LD to (2) The triangle OAB has an area of 5 cm² (b) Calculate the area of the shaded region. cm² (3)

YOUR SOLUTION

02 Use equal parallel vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q25

Let be the origin

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

Read the graph

$$B = \mathbf{a}, \quad D = \mathbf{b}, \quad C = \mathbf{a} + \mathbf{b}$$

Finish: $\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

25 ABCD is a parallelogram and ADM is a straight line. Diagram NOT accurately drawn A D K M B L C AB to = a BC to = b DM to = 1 2 b K is the point on AB such that $AK : AB = \lambda : 1$ L is the point on CD such that $CL : CD = \mu : 1$ KLM is a straight line. Given that $\lambda : \mu = 1 : 2$ use a vector method to find the value of λ and the value of μ $\lambda = \mu =$ O A O A S 100 A S

YOUR SOLUTION

03 Use midpoint or ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q23

Split the ratio

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

Finish: $AP : PM = 3 : 1$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

23 Diagram NOT accurately drawn 4a 2b A B C M P ABC is a triangle. The midpoint of BC is M. P is a point on AM. AB to $= 4a$ AC to $= 2b$ AP to $= 3$ $2a + 3$ $4b$ Find the ratio $AP : PM$

YOUR SOLUTION

04 Prove collinearity

TRIGGER *"show that", "prove", "give a reason"*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q26

Read the graph

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

Read the graph

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

Finish: $\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Use vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q23

Use vectors

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

Find the midpoint

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

Finish: $OP : PM = 4 : 1$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q26

Read the graph

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

Read the graph

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

Finish: $\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

26 The diagram shows trapezium OACB. Diagram NOT accurately drawn A B C O
4b 6b 3a OA to = 3a OB to = 6b AC to = 4b N is the point on OC such that ANB is a
straight line. Find ON to as a simplified expression in terms of a and b.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Follow the route
"work out", "show", "hence", a familiar exam phrase	Use equal parallel vectors
"work out", "show", "hence", a familiar exam phrase	Use midpoint or ratio
"show that", "prove", "give a reason"	Prove collinearity
"work out", "show", "hence", a familiar exam phrase	Use vectors
"work out", "show", "hence", a familiar exam phrase	Read the graph
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Transformations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

TRANSFORMATIONS · CH5 · L15



ENG. ESLAM AHMED

The Map

This lesson collapses Transformations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Identify the transformation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use centre and scale factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Write the vector translation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Describe a combined movement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use Transformations tag correct

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 10 website questions, 10 checked solutions, 7 local PDFs read.

01 Identify the transformation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{new point} = (1,2) + \frac{1}{2}(\text{old point} - (1,2))$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q8

Use

$$\text{new point} = (1,2) + \frac{1}{2}(\text{old point} - (1,2))$$

The vertices

$$(7,10), (15,10), (15,4), (11,4)$$

Finish: draw the enlarged shape with vertices $(4,6), (8,6), (8,3), (6,3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use centre and scale factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{new point} = (1,2) + \frac{1}{2} (\text{old point} - (1,2))$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q8

Use

$$\text{new point} = (1,2) + \frac{1}{2} (\text{old point} - (1,2))$$

The vertices

$(7,10), (15,10), (15,4), (11,4)$

Finish: draw the enlarged shape with vertices $(4,6), (8,6), (8,3), (6,3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Write the vector translation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 3

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q1

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale factor is

3

Finish: (a) enlargement, scale factor 3, centre (3,0); (b) vertices (1,4), (2,4), (1,6).

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

Answer ALL TWENTY FIVE questions. Write your answers in the spaces provided.
You must write down all the stages in your working. 1 O x y 10 8 6 4 2 A B 2 4 6 8 10 12
14 (a) Describe fully the single transformation that maps triangle A onto triangle B (3)
(b) On the grid above, translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ Label your triangle C (1)

YOUR SOLUTION

04 Describe a combined movement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(x, y) \mapsto (-6 - x, 4 - y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q5

(a) For a rotation about

$$(x, y) \mapsto (-6 - x, 4 - y)$$

The image has vertices

$$(-3, 9), (-9, 9), (-9, 6), (-6, 6)$$

Finish: (a) vertices $(-3, 9), (-9, 9), (-9, 6), (-6, 6)$; (b) vertices $(7, 3), (13, 3), (13, 6), (10, 6)$; (c) enlargement, scale factor 2, centre $(-3, 3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 Here is a shape P drawn on a grid of squares. x y 2 4 6 8 2 4 6 8 10 12 14 O -2 -4 -6
-8 -2 -4 -6 -8 -10 P 10 (a) On the grid, rotate shape P 180 degrees about the point (-3, 2)
Label the new shape Q. (2) (b) On the grid, translate shape P by the vector 10 8 Label
the new shape R. (1) Here are triangle T and triangle U drawn on a grid of squares.
x y 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9 10 11 12 O -1 -2 -3 -4 T U (c) Describe fully the single
transformation that maps triangle T onto triangle U. (3)

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(x, y) \mapsto (-6 - x, 4 - y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q5

(a) For a rotation about

$$(x, y) \mapsto (-6 - x, 4 - y)$$

The image has vertices

$$(-3, 9), (-9, 9), (-9, 6), (-6, 6)$$

Finish: (a) vertices $(-3, 9), (-9, 9), (-9, 6), (-6, 6)$; (b) vertices $(7, 3), (13, 3), (13, 6), (10, 6)$; (c) enlargement, scale factor 2, centre $(-3, 3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

5 Here is a shape P drawn on a grid of squares. x y 2 4 6 8 2 4 6 8 10 12 14 O -2 -4 -6
-8 -2 -4 -6 -8 -10 P 10 (a) On the grid, rotate shape P 180 degrees about the point (-3, 2)
Label the new shape Q. (2) (b) On the grid, translate shape P by the vector 10 8 Label
the new shape R. (1) Here are triangle T and triangle U drawn on a grid of squares.
x y 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9 10 11 12 O -1 -2 -3 -4 T U (c) Describe fully the single
transformation that maps triangle T onto triangle U. (3)

YOUR SOLUTION

06 Use Transformations tag correct

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 3

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q1

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale factor is

3

Finish: (a) enlargement, scale factor 3, centre (3,0); (b) vertices (1,4), (2,4), (1,6).

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

Answer ALL TWENTY FIVE questions. Write your answers in the spaces provided.
You must write down all the stages in your working. 1 O x y 10 8 6 4 2 A B 2 4 6 8 10 12
14 (a) Describe fully the single transformation that maps triangle A onto triangle B (3)
(b) On the grid above, translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ Label your triangle C (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Identify the transformation
"work out", "show", "hence", a familiar exam phrase	Use centre and scale factor
"work out", "show", "hence", a familiar exam phrase	Write the vector translation
"work out", "show", "hence", a familiar exam phrase	Describe a combined movement
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use Transformations tag correct

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.